

**UNIVERSITY OF BRISTOL**

**Summer Exam Period 2023 Examination Period**

**FACULTY OF ENGINEERING**

**Second Year Examination for the Degree of  
Bachelor of Science and Master of Engineering**

**COMS20011  
Data-Driven Computer Science**

**TIME ALLOWED:  
2 Hours**

**Answers to COMS20011: Data-Driven Computer Science**

**Intended Learning Outcomes:**

**Q1.** Consider the pixel values of a small image below:

|    |   |   |    |   |   |
|----|---|---|----|---|---|
| 10 | 2 | 2 | 2  | 2 | 2 |
| 10 | 2 | 1 | 14 | 2 | 2 |
| 10 | 2 | 2 | 13 | 0 | 2 |
| 10 | 2 | 1 | 15 | 2 | 2 |
| 10 | 2 | 2 | 15 | 1 | 2 |
| 10 | 2 | 2 | 2  | 2 | 2 |

Using the position of the pixel with value 13 as the centre pixel to be convolved, apply the following convolution filter once using 8-connectivity and once using 4-connectivity:

$$\begin{pmatrix} -1 & 1 & -1 \\ -1 & 3 & -1 \\ -1 & 1 & -1 \end{pmatrix}$$

Which of the options below is the correct answer for the new pixel value in each connectivity case?

- A. For 8-connectivity it is 76, and for 4-connectivity it is 70
- B. For 8-connectivity it is 60, and for 4-connectivity it is 66**
- C. For 8-connectivity it is 60, and for 4-connectivity it is 76
- D. For 8-connectivity it is 66, and for 4-connectivity it is 60
- E. For 8-connectivity it is 70, and for 4-connectivity it is 76

[6 marks]

**Solution:** B - convolve or in fact correlate (since the filter is symmetric) with 8 neighbours or 4 neighbours.

**Q2.** REDACTED

**Q3.** Naive Bayes Classifier - The table below shows the probability of certain words from amongst a large selection of spam and not spam emails received at a university. The occurrence of the words and their probabilities are independent of each other.

| Word    | $p(\text{word} \text{spam})$ | $p(\text{word} \neg\text{spam})$ |
|---------|------------------------------|----------------------------------|
| Ink     | 0.80                         | 0.30                             |
| Term    | 0.02                         | 0.93                             |
| Summer  | 0.40                         | 0.65                             |
| Printer | 0.18                         | 0.75                             |
| Bulk    | 0.70                         | 0.10                             |

(cont.)

Making a Naive Bayes assumption, compute the probability of sentence  $S1$  below being spam and the probability of sentence  $S2$  below not being spam:

$S1$ - Buy printer ink in bulk at prices seen last Summer.

$S2$ - The Summer term notes are by the printer.

Choose the correct option for  $P(S1|spam)$  and  $P(S2|not\ spam)$ :

- A.  $P(S1|spam) = 0.0403$  and  $P(S2|not\ spam) = 0.0146$
- B.  $P(S1|spam) = 0.0146$  and  $P(S2|not\ spam) = 0.4534$
- C.  $P(S1|spam) = 0.0403$  and  $P(S2|not\ spam) = 0.0014$
- D.  $P(S1|spam) = 0.0146$  and  $P(S2|not\ spam) = 0.4095$
- E.  $P(S1|spam) = 0.0403$  and  $P(S2|not\ spam) = 0.4534$**

[6 marks]

**Solution:** E -  $P(S1|spam) = 0.18 \times 0.80 \times 0.70 \times 0.40 = 0.0403$  and  $P(S2|not\ spam) = 0.65 \times 0.93 \times 0.75 = 0.4095$

**Q4.** For a digitised sample acquired using 4Hz sampling and 4 quantisation levels, the following file has been provided:

001010110101001001101011101010

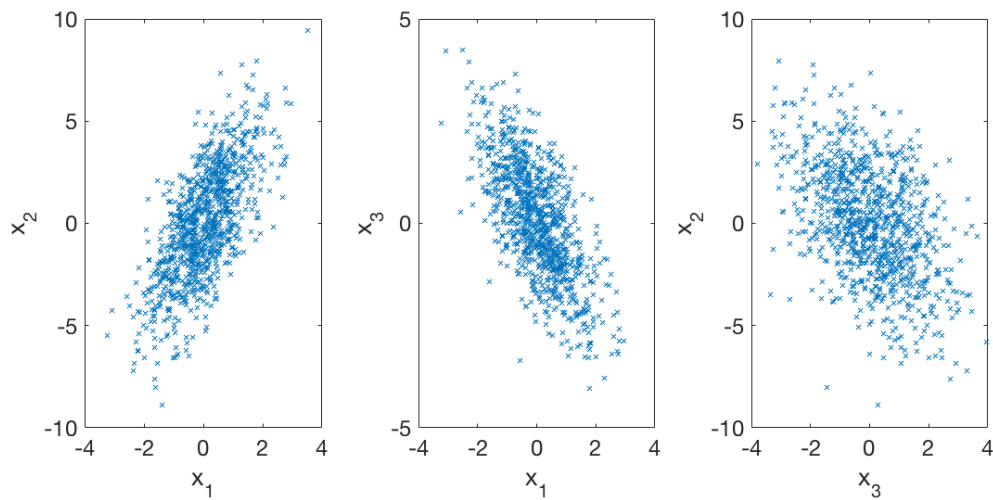
The first sample collected after the first second of recording has passed is equal to:

- A. 0110
- B. 1011
- C. 01**
- D. 10
- E. 0010

[3 marks]

**Solution:** C - 4 quantisation levels requires 2 binary digits. 4Hz is 1/4 seconds, then at 1 second we have 00101011, and the first sample collected next is 01.

**Q5.** For three-dimensional data  $X = (x_1, x_2, x_3)$ , we plot each variable against the other as shown below:



Given these plots, determine which of the following is a reasonable estimate of the covariance matrix  $\Sigma$  of dataset  $X$ ?

**A.**  $\Sigma = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 7 & -2 \\ -1 & -2 & 2 \end{bmatrix}$

B.  $\Sigma = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 1 & -2 \\ -1 & -2 & 7 \end{bmatrix}$

C.  $\Sigma = \begin{bmatrix} 1 & -2 & 1 \\ -2 & 1 & 2 \\ 1 & 2 & 7 \end{bmatrix}$

D.  $\Sigma = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 7 & -2 \\ -1 & -2 & 2 \end{bmatrix}$

E.  $\Sigma = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 7 & -2 \\ -1 & -2 & 0 \end{bmatrix}$

[5 marks]

**Solution:** A

**Q6.** Which of the following 2D matrices are NOT separable? Ignore normalisation factors which are not stated here.

$$M_1 = \begin{pmatrix} -1 & 3 & -1 \\ -3 & 9 & -3 \\ 1 & -3 & 1 \end{pmatrix}$$

$$M_2 = \begin{pmatrix} 1 & 0 & -1 \\ 2 & 0 & -2 \\ -1 & 0 & 1 \end{pmatrix}$$

$$M_3 = \begin{pmatrix} -1 & 4 & -1 \\ -1 & 8 & -1 \\ -1 & 4 & -1 \end{pmatrix}$$

$$M_4 = \begin{pmatrix} 1 & 2 & -1 & 2 & 4 \\ 2 & 4 & -2 & 4 & 8 \\ -1 & -2 & 1 & -2 & -4 \\ 2 & 4 & -2 & 4 & 8 \\ 4 & 8 & -4 & 8 & 16 \end{pmatrix}$$

$$M_5 = \begin{pmatrix} 1 & 0 & 0 & -1 \\ 2 & 1 & 1 & -2 \\ 1 & 0 & 0 & -1 \\ 2 & 1 & 1 & -2 \end{pmatrix}$$

Choose the correct option:

- A.  $M_1$  and  $M_5$
- B.  $M_2$  and  $M_3$  and  $M_5$
- C.  $M_1$  and  $M_4$
- D.  $M_2$  and  $M_3$  and  $M_4$
- E.  $M_3$  and  $M_5$**

[5 marks]

**Solution:** E - Only  $M_3$  and  $M_5$  are not separable - the others can be arrived at with one column and one row vector. Verify using Python or Matlab.

**Q7.** Two eigenvalues of the matrix below are 1 and 8:

$$\begin{bmatrix} 1 & -2 & 0 & 5 \\ 0 & 7 & 1 & 5 \\ 0 & 4 & 4 & 0 \\ 0 & 0 & 0 & 2 \end{bmatrix}$$

What are the other two eigenvalues?

- A. 3 and 4
- B. 4 and 2
- C. 2 and 3**
- D. 1 and 3
- E. 0 and 4

[4 marks]

**Solution:** C - Sum of the variances (main diagonal elements = 14) = sum of the eigenvalues, so answer has to be 2 and 3 for all the eigenvalues to sum up to 14 too

**Q8.** REDACTED

**Q9.** The eigenvalues of a dataset are: [26.0, 16.0, 13.0, 5.0, 4.0, 3.0, 1.95, 0.85, 0.60]. Approximately what variance in the dataset do the first 4 eigenvalues represent?

- A. 91.2%
- B. 88.6%
- C. 93.3%
- D. 77.3%
- E. 85.2%**

[5 marks]

**Solution:** E - Sum of the first 4 eigenvalues divided by the sum of all the eigenvalues, multiplied by 100 and rounded to 1 decimal point.

**Q10.** What is the minimum Edit Distance between the words "Sunday" and "Saturday"?

- A. 4
- B. 6
- C. 7
- D. 3**
- E. 2

[3 marks]

**Solution:** D - We would need to convert "un" to "atur" using 3 operations: substitute 'n' with 'r', insert 'a', insert 't'

**Q11.** Which of these is NOT a potential cause of overfitting?

- A. Choosing a function class that is too complex
- B. Choosing a function class that is too simple**
- C. No regularisation.
- D. Too few datapoints.
- E. Datapoints only cover a small region in the input space.

[5 marks]

**Q12.** Which of these statements about cross-validation is FALSE:

- A. Cross-validation can be used to assess overfitting.
- B. Cross-validation reports performance on the training data used to fit the function.**
- C. Cross-validation can be used to choose the function class.
- D. Cross-validation can be used to choose the amount of regularisation.
- E. Cross-validation can be computationally expensive if we have more than one or two hyperparameters.

[5 marks]

**Q13.** Which of these statements about the logarithm and its use in data-science is FALSE:

- A. The logarithm converts products into sums, i.e.  $\log ab = \log a + \log b$ .
- B. The logarithm converts powers into products, i.e.  $\log a^b = b \log a$ .
- C. The gradient of the logarithm is  $\frac{\partial \log p}{\partial p} = 2p^{-1}$ .**
- D. Using log-probabilities rather than “raw” probabilities helps us avoid numerical under/overflow.
- E. When doing maximum-likelihood fitting, the parameters with the highest log-likelihood are the same as the parameters with the highest “raw” likelihood.

[5 marks]

**Q14.** REDACTED

**Q15.** For the data in the table, fit a model of the form  $\hat{y} = w_1 + w_2x$

| x | y    |
|---|------|
| 0 | -4.2 |
| 1 | -2.3 |
| 2 | -0.1 |
| 3 | 2.1  |
| 4 | 3.9  |

[5 marks]

- A.  $w_1 = -4.20$ ,  $w_2 = 2.12$ .
- B.  $w_1 = -4.24$ ,  $w_2 = 2.06$ .**
- C.  $w_1 = -4.30$ ,  $w_2 = 2.12$ .
- D.  $w_1 = -4.32$ ,  $w_2 = 2.06$ .
- E.  $w_1 = -4.35$ ,  $w_2 = 2.12$ .

**Q16.** For the data in the table, fit a model of the form  $\hat{y} = w_1x + w_2x^2$

(cont.)

| x | y    |
|---|------|
| 0 | -4.2 |
| 1 | -2.3 |
| 2 | -0.1 |
| 3 | 2.1  |
| 4 | 3.9  |

[5 marks]

- A.  $w_1 = -1.42, w_2 = 0.603.$
- B.  $w_1 = -1.55, w_2 = 0.654.$
- C.  $w_1 = -1.60, w_2 = 0.674.$**
- D.  $w_1 = -1.78, w_2 = 0.687.$
- E.  $w_1 = -1.82, w_2 = 0.742.$

**Q17.** For the data in the table, fit a model of the form  $\hat{y}_i = w_1 X_{i1} + w_2 X_{i2}$

| $X_{i1}$ | $X_{i2}$ | $y_i$ |
|----------|----------|-------|
| -1       | -1       | -2.6  |
| -1       | 1        | -0.2  |
| 1        | -1       | 0.5   |
| 1        | 1        | 2.4   |

[5 marks]

- A.  $w_1 = 1.275, w_2 = 0.925$
- B.  $w_1 = 1.305, w_2 = 0.970$
- C.  $w_1 = 1.350, w_2 = 1.025$
- D.  $w_1 = 1.400, w_2 = 1.505$
- E.  $w_1 = 1.425, w_2 = 1.075$**

**Q18.** REDACTED

**Q19.** We have  $N$  datapoints,  $x_1, \dots, x_N$ , distributed according to,

$$P(x_i|\mu) \propto \frac{1}{x_i} e^{-(\log x_i - \mu)^2/2}$$

What is the maximum-likelihood solution for  $\mu$ ?

- A.  $\mu = \frac{1}{N} \sum_{i=1}^N x_i.$
- B.  $\mu = \frac{1}{2N} \sum_{i=1}^N \log x_i$
- C.  $\mu = \frac{1}{N} \sum_{i=1}^N \log x_i$**
- D.  $\mu = \frac{1}{2N} \sum_{i=1}^N e^{x_i}$
- E.  $\mu = \frac{1}{N} \sum_{i=1}^N e^{x_i}$

[5 marks]

**Solution:**

$$\log P(x|\mu) = \sum_i (-\log x_i - (\log x_i - \mu)^2/2) \quad (1)$$

$$\log P(x|\mu) = \sum_i (-\log x_i - (\log x_i)^2/2 + \mu \log x_i - \mu^2/2) \quad (2)$$

$$0 = \frac{\partial}{\partial \mu} \log P(x|\mu) = \sum_i (\log x_i - \mu) \quad (3)$$

$$0 = \sum_i (\log x_i) - N\mu \quad (4)$$

$$N\mu = \sum_i (\log x_i) \quad (5)$$

$$\mu = \frac{1}{N} \sum_i (\log x_i) \quad (6)$$

**Q20.** We have  $N$  datapoints,  $x_1, \dots, x_N$ , distributed according to,

$$P(x_i|\beta) \propto \beta^2 \frac{1}{x_i^3} e^{-\beta/x_i}$$

What is the maximum-likelihood solution for  $\beta$ ?

A.  $\frac{1}{2} \left( \frac{1}{N} \sum_{i=1}^N x_i \right)$

B.  $\frac{1/2}{\frac{1}{N} \sum_{i=1}^N x_i}$

C.  $\frac{1/2}{\frac{1}{N} \sum_{i=1}^N (1/x_i)}$

D.  $\frac{2}{\frac{1}{N} \sum_{i=1}^N x_i}$

**E.**  $\frac{2}{\frac{1}{N} \sum_{i=1}^N (1/x_i)}$

[5 marks]

(cont.)

**Solution:**

$$\log P(x|\beta) = \sum_{i=1}^N (2 \log \beta - 3 \log x - \beta/x_i) \quad (7)$$

$$= N(2 \log \beta - 3 \log x) - \beta \sum_{i=1}^N (1/x_i) \quad (8)$$

$$0 = \frac{\partial}{\partial \beta} \log P(x|\beta) = 2N \frac{1}{\beta} - \sum_{i=1}^N (1/x_i) \quad (9)$$

$$\sum_{i=1}^N (1/x_i) = 2N \frac{1}{\beta} \quad (10)$$

$$\beta = \frac{2N}{\sum_{i=1}^N (1/x_i)} \quad (11)$$

$$\beta = \frac{2}{\frac{1}{N} \sum_{i=1}^N (1/x_i)} \quad (12)$$